

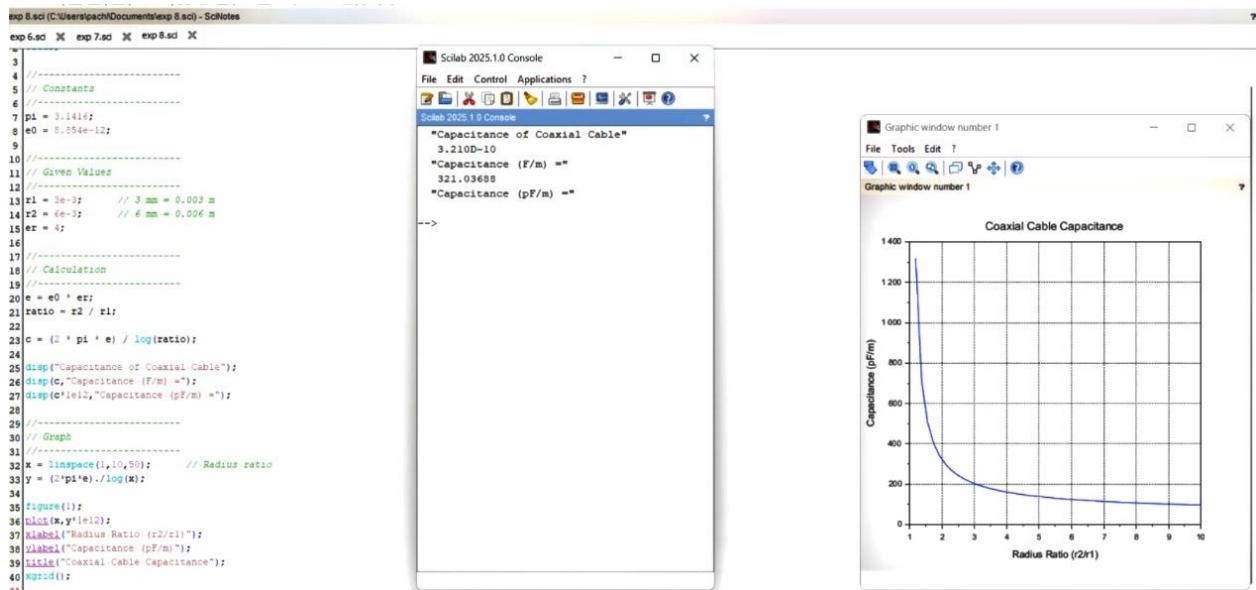
Exp no:8 -Determination and plotting of Capacitance of Co-axial cable per unit length with different (i) inner radius, (ii) outer radius and (iii) relative permittivity of the dielectric medium.

Aim: To write program for Capacitance of co-axial cable per unit length using

SCILAB. Software required:

- SCILAB version 6.0.1

Program:



Calculation:

$\epsilon_r = 8$

radius 1 = 3 mm

radius 2 = 6 mm

$\epsilon_1 = 4$

Relative permittivity

$$\epsilon = \epsilon_0 \times \epsilon_r$$

$$= (8.854 \times 10^{-12}) \times 4$$

$$= 3.5416 \times 10^{-11} \text{ F/m}$$

Radius ratio

$$n = \frac{r_2}{r_1} = \frac{0.006}{0.003} = 2$$

Natural logarithm

$$\ln(2) = 0.6931$$

capacitance per unit

$$C = \frac{2\pi (3.5416 \times 10^{-11})}{0.6931}$$

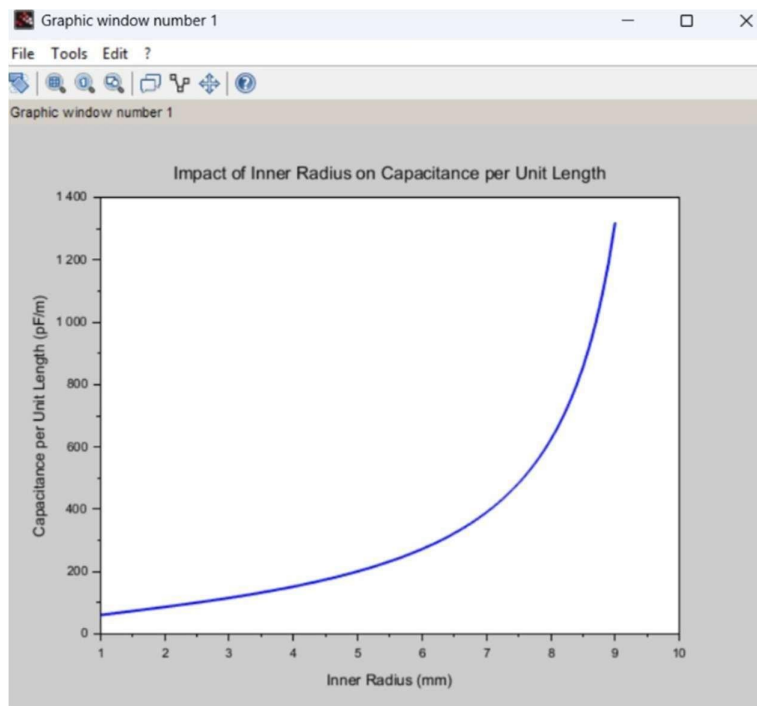
$$= 3.21 \times 10^{-10} \text{ F/m}$$

$$C = 321 \text{ pF/m}$$

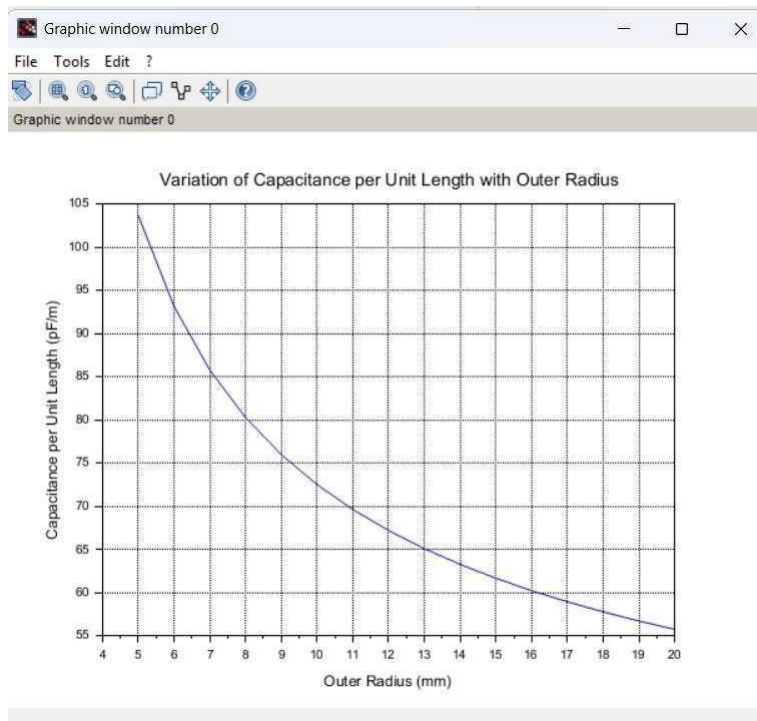
CASE STUDY 1: Variation with Inner Radius

exp 5.sci X exp 5 case 1.sci X exp 5 case II.sci X exp 5 case III.sci X exp 3.sci X exp

```
1 clc;
2 clear;
3 r1=input("Enter the value of radius1=");
4 r2=input("Enter the value of radius2=");
5 er=input("Enter the value of er=");
6
7 r=r2/r1;
8 e0=8.854*10^-12;
9 pi=3.14;
10 e=e0*er;
11
12 c=(2*pi*e)/(log(r));
13 disp('THE INNER AND OUTER RADIUS OF COAXIAL CABLE=',[c]);
14
15 x=linspace(0,10,50);
16 y=c*exp(-0.1*x);
17
18 figure();
19 subplot(1,1,1);
20 plot2d3(x,y);
21 xlabel("R");
22 ylabel("Capacitance");
23 title("INNER AND OUTER RADIUS");
24
25 |
```



CASE STUDY 2: Variation with Outer Radius

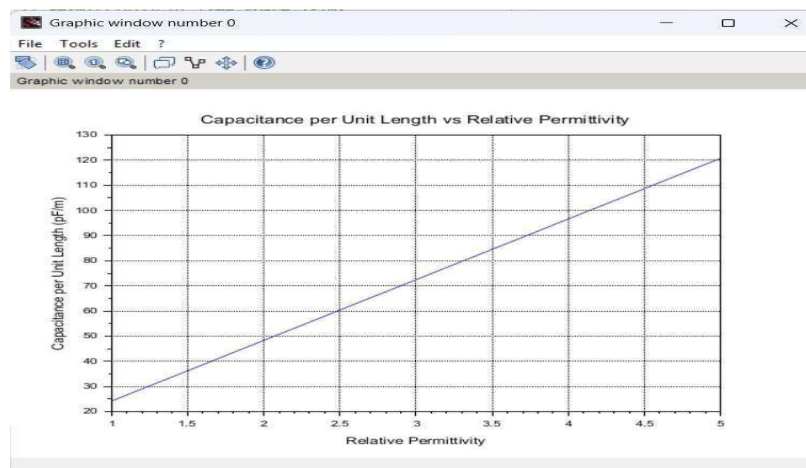


CASE STUDY 3: Variation with Relative Permittivity

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*exp 7.sci  exp6 case i.sci  exp6 caseii.sci  exp6 caseiii.sci  exp7 c1.sci  exp c2.sci  exp8
1
2 clc;
3 clear;
4
5 //Constants
6 epsilon0 = 8.854e-12; //Permittivity of free space (F/m)
7 epsilon_r = 3.0; //Relative permittivity
8 a = 1e-3; //Inner radius (1 mm)
9
10 //Outer radius values (5 mm to 20 mm)
11 b_values = 5e-3:1e-3:20e-3;
12
13 //Initialize capacitance array
14 C_prime = zeros(b_values);
15
16 //Calculate capacitance per unit length
17 for i = 1:length(b_values)
18     b = b_values(i);
19     epsilon = epsilon0 * epsilon_r;
20     C_prime(i) = (2 * pi * epsilon) / log(b / a);
21 end
22
23 //Plot
24 scf(0);
25 plot(b_values * 1e3, C_prime * 1e12);
26
27 xtitle("Variation of Capacitance per Unit Length with Outer Radius", ...
28        "Outer Radius (mm)", ...
29        "Capacitance per Unit Length (pF/m)");
30
31 xgrid();
32

```



Result : Thus the program was verified for Capacitance of co-axial cable per unit length using SCILAB was successfully.

